Project 3

Part 1:  Harris Corner Detector

Below is the original image used in the experiment:

1. Why is Gaussian smoothing applied to the gradient products and what effect does it have?

Gaussian smoothing is applied to reduce noise in the image. Raw gradients react strongly to small intensity fluctuations, which can produce many false corners. By smoothing the gradient products, the values become more stable and more representative of the local structure.

As a result, the detector becomes more reliable and detects true corners while ignoring noise.

2. What would happen if the structure used by the detector did not include the term that combines the horizontal and vertical gradients?

The missing term describes how intensity changes in the horizontal direction relate to intensity changes in the vertical direction. This relationship is essential for identifying corners.

If only the individual horizontal and vertical variations were kept, the detector would still recognize edges, but it would fail to detect real corners. Many important points would be missed because a corner requires significant changes in both directions at the same location.

Below are the gradient images before smoothing:

3. How does the Harris response indicate the strength of a corner, and what happens if the threshold is set too high or too low?

The Harris response becomes large when the image has strong intensity changes in two different directions at once — which is the definition of a corner.

- If the threshold is too high, only very strong corners will be detected, and many real but weaker corners will be ignored.

- If the threshold is too low, many irrelevant points or noise patterns will be incorrectly marked as corners.

Below is the Harris response (R map):

4. What is the role of the constant k, and how does changing it affect corner detection?

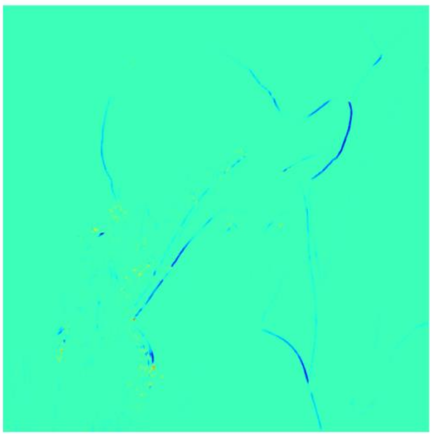
The constant k determines how strict the detector is when distinguishing between corners and edges.

- If k is too small, the detector becomes overly sensitive and marks too many points as corners, including false positives.

- If k is too large, the detector becomes too strict and may miss real corners that are not extremely strong.

Choosing k is a balance between sensitivity and reliability.

Final corner detection result:



5. Harris Corner Detector: Properties

A. Is it robust in the presence of noise?

Yes. Because the method includes Gaussian smoothing, the detector becomes resistant to random variations and noise in the image.
This makes it stable and reliable even when the image is not perfectly clean.

B. Is it invariant to scale or rotation?

- Rotation: Yes. If the image is rotated, the same corner will still be detected because the local intensity structure remains similar under rotation.

- Scale: No. If the image is resized, corners change their size and the detector does not adapt to these changes.
  Methods like SIFT or SURF are used when scale invariance is required.

C. Is it sensitive to illumination changes?

Yes. Since the method relies on intensity variations, strong illumination changes can affect gradient values and therefore influence the detection results.
Uniform lighting changes can reduce the detector's performance.

D. Can it handle affine transformations or perspective changes?

No. The detector works reliably only when the viewpoint is similar.
Under perspective distortion (for example, viewing the same object from an angle), the shape of corners changes and the method fails to identify them correctly.
More advanced methods such as Harris-Affine, SIFT, or ORB are needed for these cases.

Final result – detected corners:

Part 2: LoG Blob Detector

6. Purpose of the Laplacian of Gaussian (LoG) Filter in Blob Detection

The Laplacian of Gaussian (LoG) filter is used to enhance and identify blob-like structures in an image. By combining Gaussian smoothing with the Laplacian operator, the filter highlights regions where the intensity changes significantly in a spatially symmetric manner. In the implemented detector, LoG is applied at multiple scales, and blob locations are determined as extrema in the scale-space representation.

The effect of the LoG filter can be observed in the detection results. For example, in the case of *butterfly.jpg*, the algorithm identifies a large number of blob structures corresponding to texture patterns on the wings.

7. Effect of Sigma and the Importance of Multiple Scales

The parameter sigma (σ) controls the spatial extent of the LoG filter. A small σ responds to fine details and small blobs, whereas a larger σ responds to coarse structures and larger blobs. Using a single value of σ would limit detection to a narrow size range; therefore, a multi-scale approach is required.

In the multi-scale LoG implementation, a geometric progression of σ values is used to build a scale-space volume. Extrema across this volume allow detection of blobs of varying sizes and enable approximate scale invariance.

This behaviour is illustrated in the results for *sunflowers.jpg*, where both small distant flower centers and larger nearby ones are successfully identified.

8. LoG Blop Detector:

A. Robustness of the LoG Blob Detector to Noise

The LoG detector includes Gaussian smoothing, which provides some degree of robustness to noise. However, noise or high-frequency texture can still cause numerous spurious responses. This is notable in *lenna.png*, where textured areas such as hair generate many blob detections, despite the pruning stage.

B. Scale and Rotation Invariance of the LoG Detector

When combined with scale normalization and multi-scale analysis, the LoG detector achieves approximate scale invariance because the response is evaluated over a range of σ values, and extrema across scales indicate blob size. The LoG filter is inherently rotation-invariant due to the isotropic nature of both the Gaussian and the Laplacian operators. Rotating the image does not alter the filter response.

This is consistent with the detection patterns observed across various images in the experiment.

C. Response of the LoG Detector to Illumination Changes

Additive changes in illumination (uniform brightening or darkening) are largely cancelled out by the second-derivative nature of the LoG filter. However, multiplicative changes in illumination or contrast affect the magnitude of the filter response. The thresholding step in the implementation determines which responses are retained as blob candidates, making the detector sensitive to global illumination variations.

This effect is observable in the differences between the results on images with strong contrast (such as *sunflowers.jpg*) versus smoother ones (such as *lenna.png*).

D. Limitations of the LoG Detector Under Affine Transformations

The LoG detector assumes that blob-like structures are circular and isotropic. Under affine or perspective transformations, circular structures become elliptical, and the standard LoG filter cannot adapt to such shape distortions. As a result, the detector is not affine-invariant.

This limitation is evident in the results on *chessboard_perspective.png*, where many false blob detections occur due to elongated and distorted patterns. Conversely, for *chessboard.jpg*, which contains no circular structures and no perspective distortion, the detector returns zero blobs as expected.